Smart Study Planner

with Priority-Based Scheduling

AP(OOP) — PCCCS495 | 4th Sem | Stream: IOT

Anushka Maitra | Roll No: 33

Problem Statement

Students and professionals often struggle to manage multiple academic deadlines with varying levels of urgency and complexity. Most existing to-do applications rely on manual prioritization and do not dynamically adjust task order when deadlines or workload conditions change. This leads to inefficient planning and missed commitments. The Smart Study Planner solves this by modeling tasks through a structured object-oriented hierarchy with a priority queue that automatically reorders tasks based on urgency, deadline, and task type — ensuring students always focus on what matters most.

System Design

Package Structure

com.studyplanner.model

Abstract + concrete task classes

com.studyplanner.manager

SchedulerManager — core logic

com.studyplanner.storage

TaskStorage — file persistence

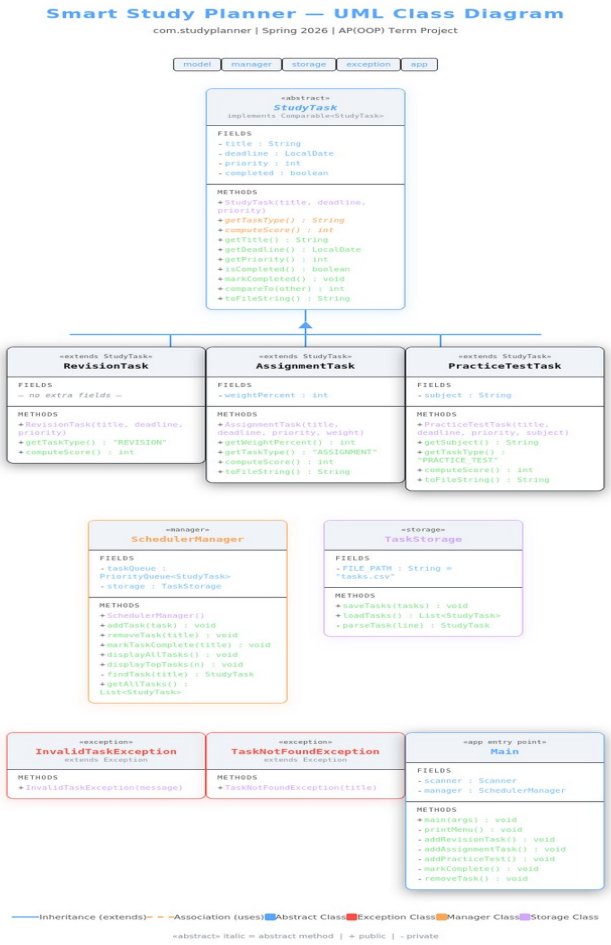
com.studyplanner.exception

Custom exception classes

com.studyplanner.app

Main — console UI entry point

UML Class Diagram



OOP Concepts Used

Abstraction

Abstract StudyTask class
forces subclasses to implement
getTaskType() + computeScore()

Inheritance

RevisionTask, AssignmentTask,
PracticeTestTask all extend
StudyTask

Polymorphism

computeScore() overridden
in each subclass — different
logic per object type

Encapsulation

All fields private
Accessed only via
public getters/setters

Exception Handling

InvalidTaskException
TaskNotFoundException
for graceful error handling

Collections

PriorityQueue auto-orders
tasks by score
O(log n) efficiency

Technical Highlight

Why PriorityQueue?

- Auto-sorts on every task insert
- No manual sorting needed
- O(log n) performance
- Works with Polymorphism via compareTo()

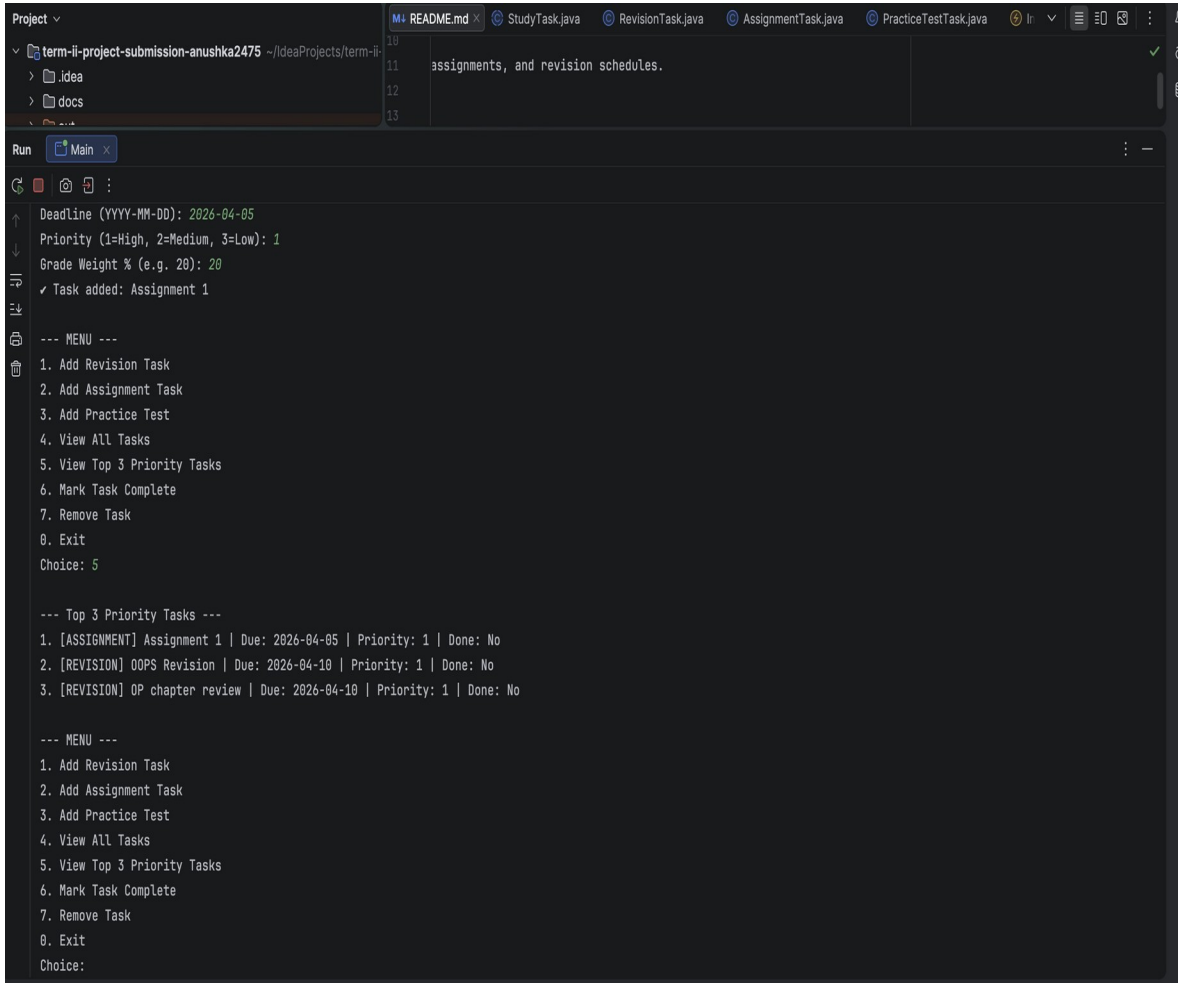
Score Formulas:

Revision: priority×10 – daysLeft

Assignment: priority×10 + weight – days

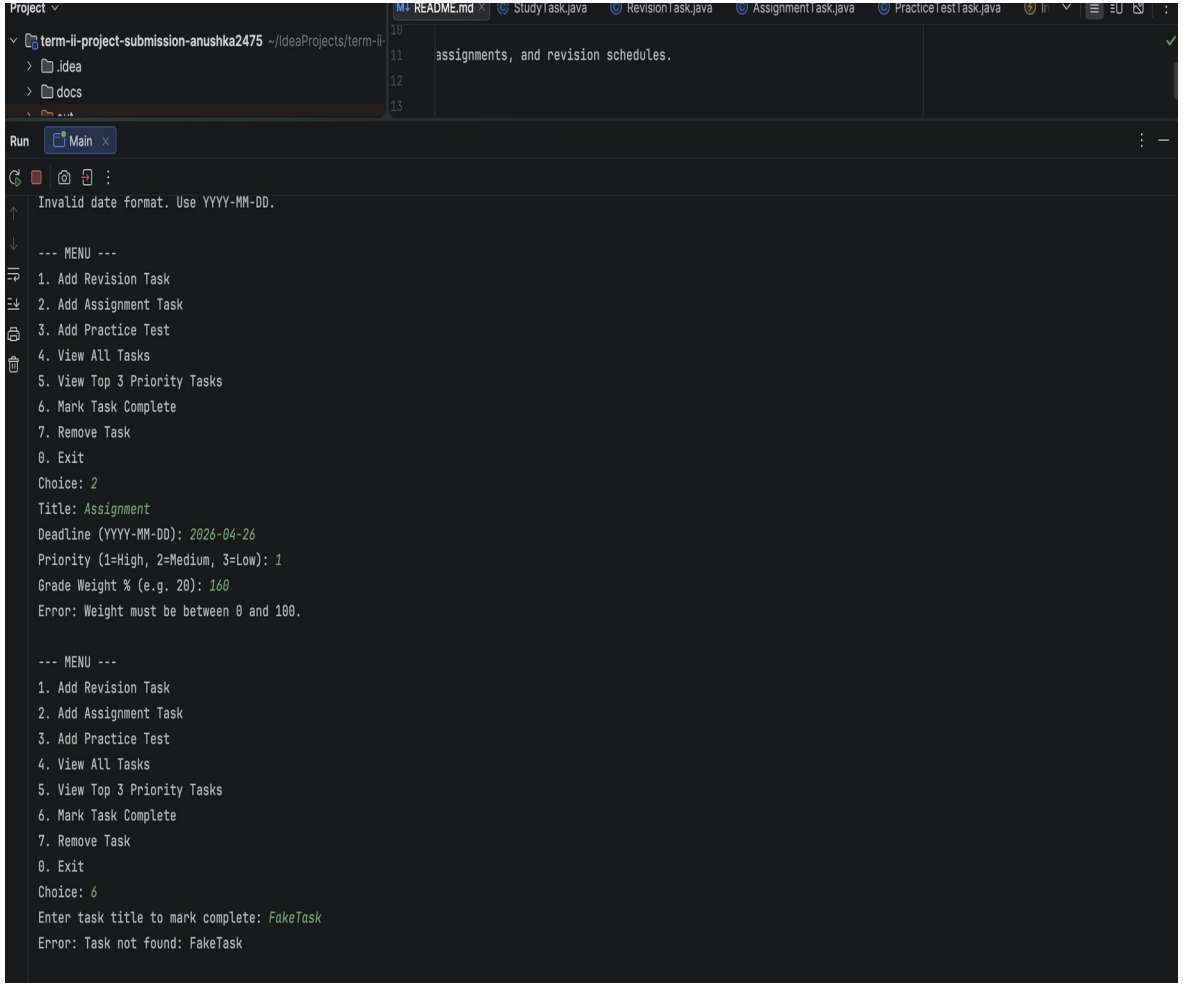
Practice: priority×15 – daysLeft×2

Program Output



Live Demo

Demo Steps

1. Run Main.java → See menu
2. Press 1 → Add Revision Task
3. Press 5 → View top 3 tasks
4. Enter invalid date → See error
5. Exit + rerun → Tasks persist

Error Handling Output